

ST. JOSEPH COLLEGE OF ENGINEERING AND TECHNOLOGY
NTA LEVEL 5 - TECHNICIAN CERTIFICATE
CONTINUOUS ASSESSMENT TEST - II - FEBRUARY-2026
EDT 05103 - APPLIED CHEMISTRY

SEMESTER: I
DURATION: 1 Hr 30 Min.

YEAR: II
MAX. MARKS: 50

Auth: 4LAZIE

PART – A (Answer all the Questions)

(7 x 2 = 14)

1. The compressive strength of cement increases from 10^2 N/m^2 when initially set to 10^7 N/m^2 after _____ days, reaching 70% of its final strength value (10^8 N/m^2) after ____ days.
2. What is a reclaimed rubber?
3. Define leaching as a concentration method in metallurgical operation.
4. What is powder metallurgy.
5. List the methods of making metal powder in powder metallurgy
6. Why concentration is an important step that we can't skip during metallurgical operation
7. Mention the metal constitutes of Amalgam Alloy

PART – B (Answer any three Questions)

(3 x 4 = 12)

8. How lime is used in mining and metallurgy
9. Explain the meaning of the following Quicklime, Hydrated lime and Dolomite lime
10. a) Define adhesive
b) what is bonding in adhesive?
11. Explain Calcination as an oxidation method in metallurgy.
12. What is the best method for extraction of sulphide ores? Explain it in brief.
13. a) define rubber and Latex
b) Give any three defects of natural rubber.

PART – C (Answer any two Questions)

(2 x 12 = 24)

14. Clinkering stage is one of the important step in Portland cement manufacturing, systematically explain each of the process/step taking place during clinker production.
15. Aluminium has increasingly replaced iron in several modern engineering applications; however, iron/steel continue to serve as fundamental materials in engineering practice today.
a) In brief explain why aluminium has replaced iron in certain applications.
b) Discuss the limitations of aluminium that prevent it from completely replacing iron/steel in engineering application/practice today
16. After crushing/Grinding of Iron Ore (Haematite). The Next Process (Smelting) is normally carried out in a blast furnace, a tall cylindrical furnace made of steel plates lined inside with fire bricks. Describe the blast furnace as used in extraction of Iron.
17. Explain the electro-refining or hall-héroult process of extraction of Aluminium (Al) from Alumina (Al_2O_3)